

ONLINE SHOPPING BEHAVIOUR: ANALYSING CUSTOMER ACTIVITY AND PREDICTING PURCHASE INTENT USING CLASSIFICATION MODELS

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DECLARATION

I hereby declare that the project report entitled “Analysing Online Shopping Behaviour and Predicting Purchase Intent” is my original work completed as part of the IDRA Data Science and AI capstone project. The analysis, preprocessing, exploratory analysis, machine learning development, evaluation, and application work presented in this report were carried out as part of the project. External resources and software libraries used in the project have been acknowledged appropriately.

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ABSTRACT

Online shopping platforms generate large volumes of session-level behavioural data that can be used to understand customer activity and identify sessions with a higher likelihood of purchase. This project investigates whether online shopping behaviour and session information can be used to predict purchase intent. The study uses 25,000 shopping sessions and 29 original variables covering visit information, device and user characteristics, marketing channels, product information, pricing, browsing behaviour, cart activity, and purchase outcomes. The target variable is purchased, where 1 represents a purchase and 0 represents no purchase. Data-quality checks found no missing values and no duplicate rows. Leakage-prone or post-outcome variables, including revenue, payment method, cart abandonment, and review-related variables, were excluded from modelling. Categorical variables were one-hot encoded and numerical variables were standardized in a preprocessing pipeline. Logistic Regression and Random Forest were evaluated using accuracy, precision, recall, F1-score, and ROC-AUC. Random Forest achieved the strongest overall performance with 92.04% accuracy, 98.14% precision, 65.81% recall, 78.78% F1-score, and 91.31% ROC-AUC. The findings indicate that session behaviour, particularly add-to-cart activity, provides useful information for purchase prediction. The resulting model can support targeted customer engagement, subject to the limitations of the dataset and prediction-time feature availability.

Keywords: *[Online Shopping]; [Purchase Intent]; [Customer Behaviour]; [Classification]; [Random Forest]*

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1. INTRODUCTION

1.1 Background

E-commerce websites continuously record customer interactions such as pages viewed, time spent on a website, product activity, cart actions, marketing source, and purchase outcomes. Analysing these interactions can help businesses understand customer behaviour and identify patterns associated with conversion. Purchase-intent prediction is a binary classification problem in which behavioural and session attributes are used to estimate whether a shopping session will end in a purchase.

1.2 Context of the Problem

The practical stakeholders include e-commerce businesses, digital marketing teams, product teams, and customer-engagement teams. A reliable estimate of purchase likelihood can help prioritize high-intent sessions, personalize relevant offers, optimize promotional activities, and allocate marketing resources more effectively.

1.3 Scope of the Study

This study focuses on session-level online shopping behaviour and binary purchase prediction. It covers data understanding, cleaning, exploratory analysis, statistical summaries, preprocessing, classification modelling, evaluation, and a Streamlit prediction interface. It does not attempt to estimate customer lifetime value, causal effects of marketing channels, or long-term customer retention.

2. PROBLEM STATEMENT AND MOTIVATION

The central research question is: “Can we use a customer's online shopping behaviour and session information to predict whether they will make a purchase or not?” The motivation is to transform behavioural session data into an actionable prediction that can support customer engagement and conversion-oriented decision making.

3. OBJECTIVES

To understand and prepare the online shopping session dataset for analysis.

To identify behavioural patterns and relationships associated with purchase outcomes.

To identify and remove variables that could introduce post-outcome leakage.

To develop classification models for predicting whether a shopping session results in a purchase.

To evaluate and compare the models using appropriate classification metrics.

To translate analytical and modelling results into practical recommendations.

4. RESEARCH QUESTIONS

What are the major characteristics and quality conditions of the online shopping dataset?

Which behavioural and session variables show meaningful relationships with purchase outcomes?

How does purchase behaviour vary across selected customer, session, and product-related groups?

How accurately can machine learning models predict the purchase outcome?

5. DATASET DESCRIPTION AND DATA UNDERSTANDING

5.1 Data Source

The project uses the online shopping behaviour dataset supplied/used for the IDRA capstone project. The dataset file used in the project is online_shopping.csv.

5.2 Dataset Size and Structure

The raw dataset contains 25,000 observations and 29 columns. Each observation represents an online shopping session. The target variable is purchased, where 1 represents a purchase and 0 represents no purchase.

Measure	Value	Interpretation	Role
Rows	25,000	Shopping-session observations	Dataset size
Original columns	29	Customer, visit, product, behaviour and outcome fields	Dataset structure
Target	purchased	0 = no purchase; 1 = purchase	Target
No-purchase sessions	19,384 (77.54%)	Majority class	Target distribution
Purchase sessions	5,616 (22.46%)	Minority class	Target distribution

Table 1. Summary of the Dataset

5.3 Data Dictionary

Variable	Meaning	Type	Role
customer_id	Customer identifier	Integer	Excluded ID
session_id	Session identifier	Integer	Excluded ID
visit_date	Date of session	Date/String	Used for date features
device_type	Encoded device category	Integer	Feature
user_type	Encoded user category	Integer	Feature
marketing_channel	Encoded marketing source	Integer	Feature
product_id	Encoded product identifier	Integer	Categorical feature
product_category	Encoded product category	Integer	Categorical feature
unit_price	Price per unit	Float	Feature
quantity	Number of units	Integer	Feature
discount_percent	Discount percentage	Integer	Feature
discount_amount	Discount amount	Float	Feature
pages_viewed	Number of pages viewed	Integer	Feature
time_on_site_sec	Time spent on site	Integer	Feature
added_to_cart	Whether an item was added to cart	Binary	Feature
purchased	Purchase outcome	Binary	Target
rating	Recorded rating value	Integer	Feature
visit_day	Day of month	Integer	Feature

Variable	Meaning	Type	Role
customer_id	Customer identifier	Integer	Excluded ID
visit_month	Month of visit	Integer	Feature
visit_weekday	Weekday code	Integer	Feature
visit_season	Encoded season category	Integer	Feature
session_duration_bucket	Session duration category	String	Feature
location	Encoded location category	Integer	Feature

Table 2. Data Dictionary

6. DATA CLEANING

The raw dataset was inspected for missing values, duplicate records, data types, category consistency, and unusual values. No missing values and no duplicate rows were found. The `visit_date` field was converted from text to datetime before extracting calendar features. .

6.1 Missing Values

No missing values were found in any of the 29 original columns. Therefore, no imputation was required and all 25,000 observations were retained.

6.2 Duplicate Records

The duplicate-row check returned zero duplicate records. No observations were removed for duplication.

6.3 Data Types and Inconsistent Values

`visit_date` was converted to datetime using the day-month-year format present in the raw file. Calendar features were aligned with the standard weekday convention, where Monday is 0 and Sunday is 6.

6.4 Outlier Handling

Numerical ranges were inspected for variables such as `unit_price`, `discount_amount`, `pages_viewed`, and `time_on_site_sec`. No observation was removed solely because it was numerically large; values within the recorded dataset ranges were retained.

6.5 Before vs After Cleaning

Quality Measure	Before	After	Decision
Rows	25,000	25,000	No rows removed
Columns	29	29	Retained at cleaning stage
Missing values	0	0	No imputation
Duplicate rows	0	0	No duplicate removal

Table 3. Data Quality Before and After Cleaning

7. DATA PREPROCESSING

After cleaning, modelling variables were selected and the target was separated from the input features. Post-outcome or leakage-prone variables were excluded before training. The resulting modelling dataset contained 21 columns including the target and 20 input features.

7.1 Categorical Encoding

Categorical features were processed using OneHotEncoder(handle_unknown='ignore'). These included device_type, user_type, marketing_channel, product_id, product_category, added_to_cart, visit_weekday, visit_season, session_duration_bucket, and location. Product and location codes were treated as categorical labels rather than continuous measurements.

7.2 Feature Scaling and Transformation

Numerical variables were standardized using StandardScaler inside a scikit-learn ColumnTransformer. Keeping preprocessing inside the pipeline ensures consistent transformation between training and unseen test data.

7.3 Train-Test Split

The dataset was split into 80% training data and 20% testing data using random_state=42 and stratify=y. This produced 20,000 training sessions and 5,000 test sessions while preserving the target-class distribution.

7.4 Final Feature Set

Final input features: device_type, user_type, marketing_channel, product_id, product_category, unit_price, quantity, discount_percent, discount_amount, pages_viewed, time_on_site_sec, added_to_cart, rating, visit_day, visit_month, visit_weekday, visit_season, session_duration_bucket, location, and visit_year. Target: purchased.

Leakage-prone variables excluded from modelling: revenue, revenue_normalized, payment_method, cart_abandoned, review_text, and review_helpful_votes. Customer_id and session_id were also excluded as identifiers.

8. EXPLORATORY DATA ANALYSIS

8.1 Univariate Analysis

```
# Purchase distribution

plt.figure(figsize=(7, 5))

sns.countplot(data=df, x="purchased")

plt.title("Purchase Intent Distribution")
plt.xlabel("Purchase Intent (0 = No Purchase, 1 = Purchase)")
plt.ylabel("Number of Sessions")

plt.show()
```

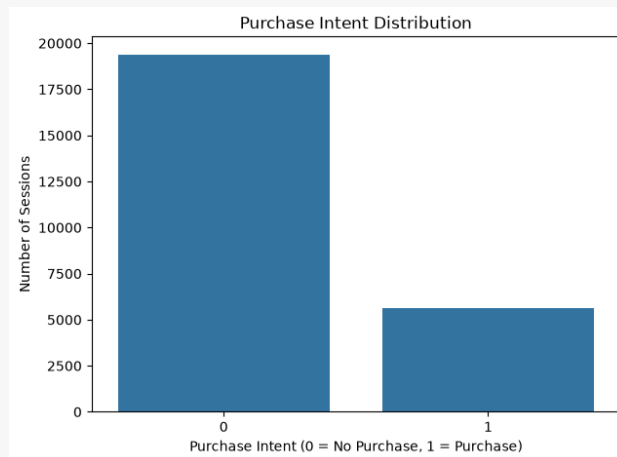


Figure 1. Distribution of [Purchase Intent]

The target distribution is imbalanced toward non-purchase sessions. There are 19,384 non-purchase sessions (77.54%) and 5,616 purchase sessions (22.46%). This imbalance was considered during model development.

8.2 Categorical Analysis

Purchase rate by device type

```
device_purchase = df.groupby("device_type")["purchased"].mean() * 100
```

```
print(device_purchase)
```

```
plt.figure(figsize=(7, 5))
```

```
device_purchase.plot(kind="bar")
```

```
plt.title("Purchase Rate by Device Type")
```

```
plt.xlabel("Device Type")
```

```
plt.ylabel("Purchase Rate (%)")
```

```
plt.xticks(rotation=0)
```

```
plt.show()
```

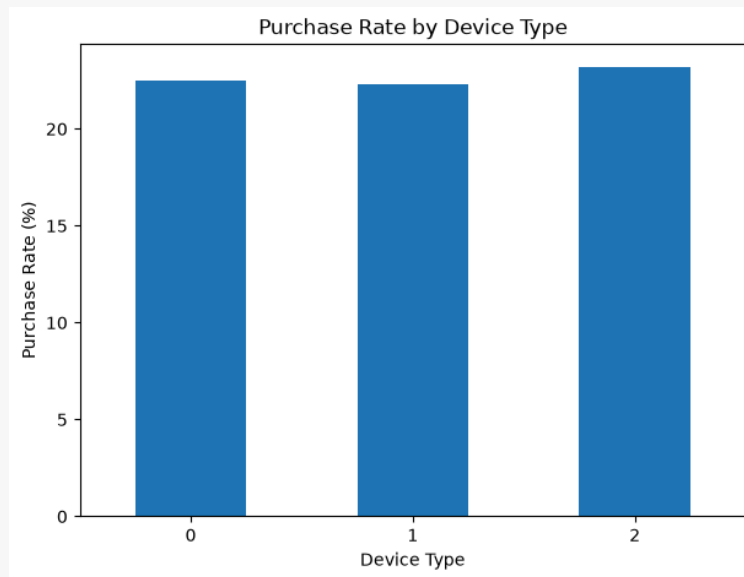


Figure 2. [Purchase Rate] Compared Across [Device Type]

Purchase rates were compared across encoded device types, user types, marketing channels, product categories, session duration groups, and other categorical variables. Numeric codes were not assigned invented business meanings because the supplied dataset did not provide a semantic mapping for the encoded categories.

8.3 Bivariate / Relationship Analysis

Purchase rate based on whether the item was added to cart

```
cart_purchase = df.groupby("added_to_cart")["purchased"].mean() * 100
```

```
print(cart_purchase)
```

```
plt.figure(figsize=(7, 5))  
cart_purchase.plot(kind="bar")
```

```
plt.title("Purchase Rate by Add-to-Cart Behaviour")  
plt.xlabel("Added to Cart (0 = No, 1 = Yes)")  
plt.ylabel("Purchase Rate (%)")  
plt.xticks(rotation=0)
```

```
plt.show()
```

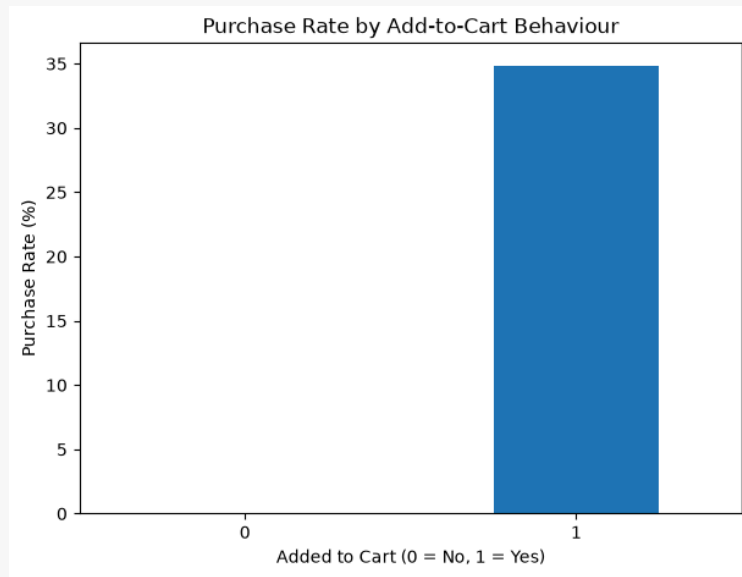


Figure 3. Relationship Between [Add-to-cart] and [Purchase Rate]

Add-to-cart activity showed a strong association with purchase. Sessions without an add-to-cart event had a 0% observed purchase rate, while sessions with an add-to-cart event had an observed purchase rate of approximately 34.8%. This is an observed association and does not by itself establish causation.

8.4 Correlation Analysis

```
correlation = df.select_dtypes(include="number").corr()["purchased"].sort_values(ascending=False)

print("Correlation with Purchase Intent:")
print(correlation)
```

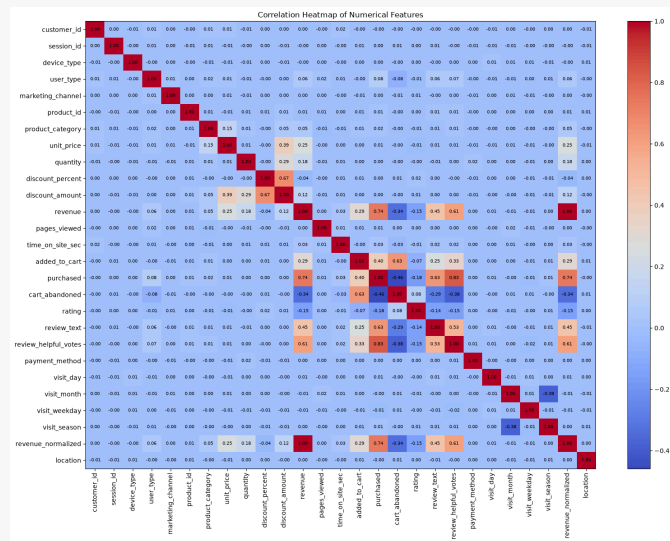


Figure 4. Correlation Heatmap of Numerical Variables

The strongest positive raw correlations with purchases were review_helpful_votes (0.8314), revenue_normalized (0.7353), revenue (0.7353), and review_text (0.6311). cart_abandoned showed a negative correlation of -0.4581. These highly associated variables were not used as model inputs when they represented post-outcome or leakage-prone information.

8.5 Other Figures

```
duration_purchase = df.groupby("session_duration_bucket")["purchased"].mean() * 100
```

```
print("Purchase Rate by Session Duration:")
print(duration_purchase)
```

```
plt.figure(figsize=(8, 5))
duration_purchase.plot(kind="bar")
```

```
plt.title("Purchase Rate by Session Duration")
plt.xlabel("Session Duration")
plt.ylabel("Purchase Rate (%)")
plt.xticks(rotation=0)
```

```
plt.show()
```

9. STATISTICAL ANALYSIS

Descriptive statistics were used to understand the scale and variability of key variables. Unit price ranged from 50.05 to 1,999.83; quantity ranged from 1 to 4; discount percentage ranged from 0 to 30; pages viewed ranged from 1 to 24; time on site ranged from 10 to 1,799 seconds; and rating ranged from 1 to 5.

Variable	Minimum	Maximum	Interpretation
unit price	50.05	1,999.83	Product price
quantity	1	4	Units
discount percent	0	30	Discount range
pages viewed	1	24	Browsing depth
time on site sec	10	1,799	Session duration

Table 4. Descriptive Statistics of Key Variables

10. FEATURE ENGINEERING

The `visit_date` field was transformed into `visit_year`, `visit_month`, `visit_day`, and `visit_weekday`. Existing date-related fields were aligned with these extracted components. No synthetic variables were added solely to increase model performance.

Feature	Source/Logic	Purpose	Type
<code>visit_year</code>	Year extracted from <code>visit_date</code>	Calendar context	Numerical
<code>visit_month</code>	Month from <code>visit_date</code>	Calendar context	Numerical
<code>visit_day</code>	Day from <code>visit_date</code>	Calendar context	Numerical
<code>visit_weekday</code>	Weekday from <code>visit_date</code>	Weekly context	Categorical

Table 5. Engineered Features

11. MODEL DEVELOPMENT

11.1 Problem Formulation

The task is binary classification. The input consists of customer, product, pricing, calendar, and session-behaviour features, while `purchased` indicates whether the session resulted in a purchase.

11.2 Model Selection

Logistic Regression was selected as an interpretable baseline. Random Forest was selected as a non-linear ensemble model capable of learning interactions among behavioural and categorical features.

11.3 Training the Model

Both models were implemented as scikit-learn pipelines combining preprocessing and model training. Logistic Regression used `max_iter=1000` and `class_weight='balanced'`. Random Forest used 200 trees, `class_weight='balanced'`, `random_state=42`, and `n_jobs=-1`.

```
from sklearn.pipeline import Pipeline
```

```
from sklearn.linear_model import LogisticRegression

# Create Logistic Regression pipeline
logistic_model = Pipeline(
    steps=[
        ("preprocessor", preprocessor),
        ("classifier", LogisticRegression(
            max_iter=1000,
            class_weight="balanced",
            random_state=42
        ))
    ]
)

# Train the model
logistic_model.fit(X_train, y_train)

print("Logistic Regression model trained successfully!")
```

12. MODEL EVALUATION

Evaluation was performed on the unseen 5,000-session test set. Accuracy measures overall correctness, precision measures the proportion of predicted purchases that were actual purchases, recall measures the proportion of actual purchases detected, F1-score balances precision and recall, and ROC-AUC measures discrimination across classification thresholds.

12.1 Classification Evaluation

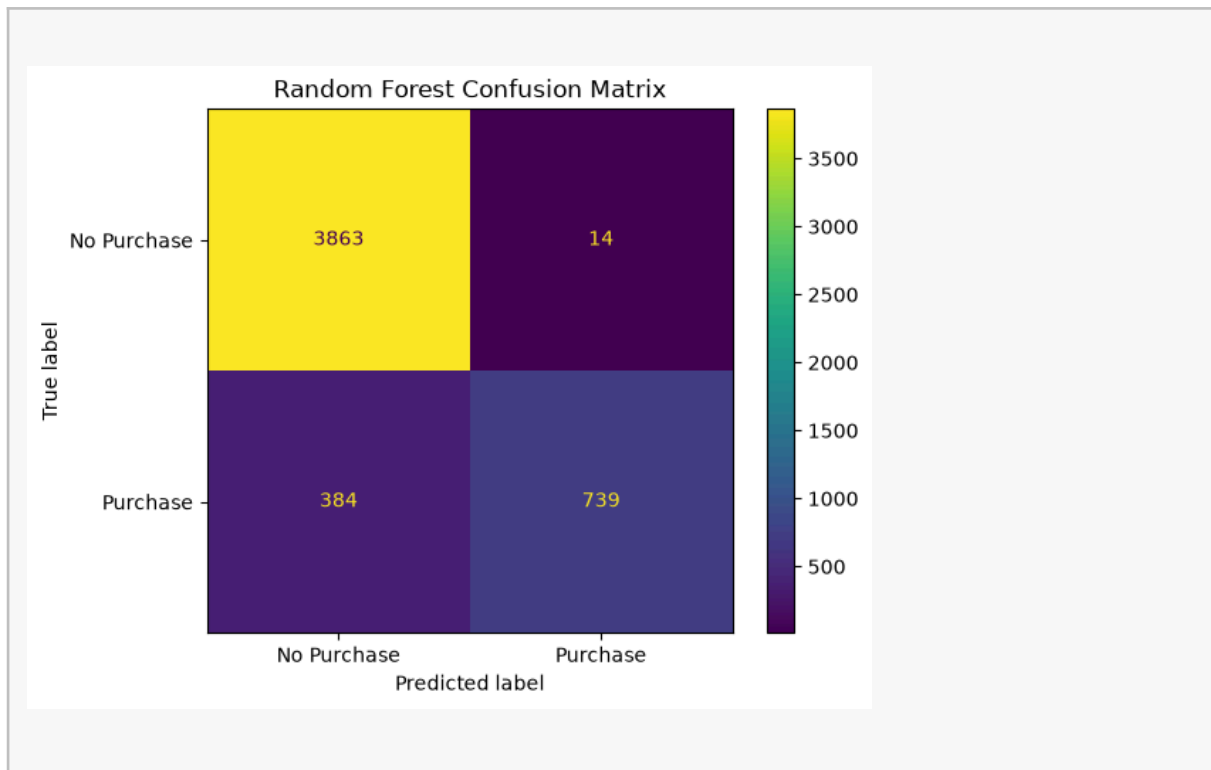


Figure 5. Confusion Matrix

Model	Accuracy	Precision	Recall
Logistic Regression	62.34%	35.40%	82.01%
Random Forest	92.04%	98.14%	65.81%

Table 7. Classification Model Performance

12.2. Overfitting and Underfitting

The available project results contain test-set performance for both models but do not include the corresponding training metrics. Therefore, the report does not claim that Random Forest is definitively free of overfitting. Cross-validation and train-versus-test comparisons would provide stronger evidence in future work.

13. RESULTS AND FINDINGS

1. Class imbalance was present in the target variable.: [Out of 25,000 sessions, **5,616 (22.46%) resulted in a purchase**, while **19,384 (77.54%) did not**. This shows that non-purchase sessions were substantially more common.]
2. Add-to-cart behaviour was strongly associated with purchase outcome. : [Sessions with `added_to_cart = 1` had a purchase rate of approximately **34.8%**, whereas sessions without adding an item to the cart had a **0% purchase rate**.]

3. Random Forest performed better than Logistic Regression. : [Random Forest achieved **92.04% accuracy, 98.14% precision, 65.81% recall, 78.78% F1-score, and 91.31% ROC-AUC**, compared with Logistic Regression's **62.34% accuracy and 75.79% ROC-AUC**.
4. The target variable was associated with some variables more strongly than others. : [The correlation analysis showed a correlation of **0.3996** between **added_to_cart** and **purchased**, while cart_abandoned had a correlation of **-0.4581** with purchased. These values indicate notable associations, although correlation does **not imply causation**.]

14. DISCUSSION

The analysis demonstrates that online shopping sessions contain useful behavioural signals for predicting purchase outcomes. Add-to-cart activity is particularly informative because it represents a deeper stage of the shopping funnel. Random Forest performed substantially better than Logistic Regression, suggesting that non-linear patterns and interactions among session variables are important for this dataset.

The Random Forest precision of 98.14% indicates that very few predicted purchases were false positives in the test set, while recall of 65.81% indicates that a meaningful number of actual purchases were still missed. The model should therefore be used with an application-specific decision threshold rather than relying only on the default classification threshold.

Observed relationships should not be interpreted as causal effects. Prediction-time feature availability must be considered before any real-world deployment.

15. LIMITATIONS

- [The dataset may not generalize to every e-commerce platform, customer population, or future time period.]
- [The evaluation uses a single train-test split rather than cross-validation.]
- [Some raw variables are post-outcome or closely connected to the outcome. Prediction-time availability must be validated carefully.]

16. CONCLUSION

This project addressed whether online shopping behaviour and session information can be used to predict purchase outcomes. A complete data science workflow was implemented, including data understanding, cleaning, preprocessing, exploratory analysis, feature engineering, classification modelling, and evaluation. Random Forest was the strongest evaluated model, achieving 92.04% accuracy, 98.14% precision, 65.81% recall, 78.78% F1-score, and 91.31% ROC-AUC on the test set. The analysis also showed a strong association between add-to-cart activity and purchase. Overall, the project demonstrates the usefulness of session-level behavioural data for purchase-intent prediction while highlighting the importance of leakage control and prediction-time feature definitions.

17. RECOMMENDATIONS AND FUTURE WORK

17.1 Recommendations

- Recommendation 1: [Use purchase-intent scores to prioritize relevant engagement for high-intent sessions.]

- Recommendation 2: [Monitor add-to-cart behaviour as a practical funnel signal because it showed a strong association with purchase.]
- Recommendation 3: [Before deployment, verify that every model feature is available at the exact prediction time.]

17.2 Future Work

Future work can include cross-validation, hyperparameter tuning, probability calibration, decision-threshold optimization, and comparison with gradient-boosting models. Time-aware validation could test generalization to future periods. A future version could also obtain documented semantic mappings for encoded categories and integrate the model with a live analytics pipeline.

18. REFERENCES

- [India Data Research Academy. (2026). IDRA Data Science and AI Capstone Project Report Template.].
- [Pedregosa, F., et al. (2011). Scikit-learn: Machine Learning in Python. Journal of Machine Learning Research, 12, 2825–2830.]
- [Harris, C. R., et al. (2020). Array programming with NumPy. Nature, 585, 357–362.]

APPENDIX A. CODE

A.1 Data Loading and Inspection

```
df = pd.read_csv("../data/online_shopping.csv")
print(df.shape)
print(df.isnull().sum())
print(df.duplicated().sum())
```

A.2 Data Cleaning

```
df["visit_date"] = pd.to_datetime(df["visit_date"], format="%d-%m-%Y")
leakage_columns = ["revenue", "revenue_normalized", "payment_method", "cart_abandoned",
"review_text", "review_helpful_votes"]
df_model = df.drop(columns=leakage_columns)
df_model = df_model.drop(columns=["customer_id", "session_id"])
```

A.3 Preprocessing

```
X = df_model.drop(columns=["purchased"])
y = df_model["purchased"]
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.20, random_state=42, stratify=y)
preprocessor = ColumnTransformer([...])
```

A.4 EDA and Visualization

```
purchase_counts = df["purchased"].value_counts()
cart_purchase = df.groupby("added_to_cart")["purchased"].mean() * 100
device_purchase = df.groupby("device_type")["purchased"].mean() * 100
correlation = df.select_dtypes(include="number").corr()["purchased"].sort_values(ascending=False)
```

A.5 Model Training and Evaluation

```
random_forest_model = Pipeline(steps=[("preprocessor", preprocessor), ("classifier",
RandomForestClassifier(n_estimators=200, random_state=42, class_weight="balanced", n_jobs=-1))]
random_forest_model.fit(X_train, y_train)
y_pred_rf = random_forest_model.predict(X_test)
y_pred_proba_rf = random_forest_model.predict_proba(X_test)[: , 1]
```

APPENDIX B. ADDITIONAL FIGURES AND TABLES

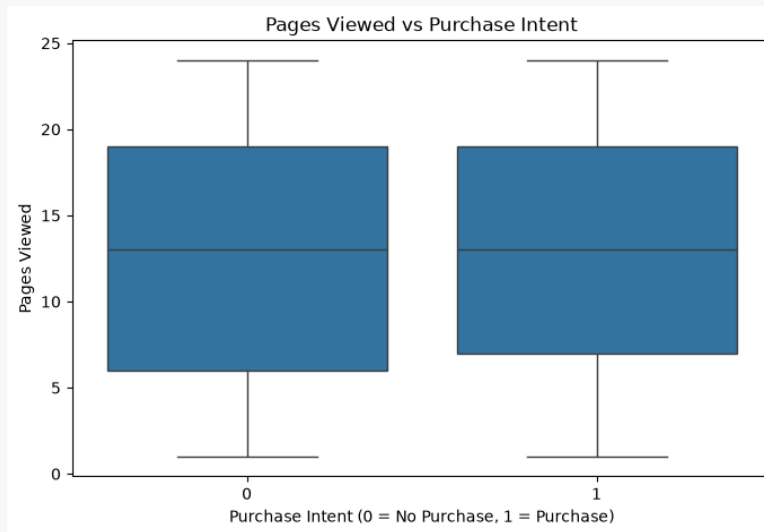


Figure 6. Purchase Intent vs Pages Viewed

Purchased	pages viewed	Time on site sec	Added to cart
0	12.512846	895.605809	0.541735
1	12.615207	929.691952	1.000000

Table 8. Additional Results Table

